

Introduction

Internet of Things
LABORATORY



CS5055 Internet of Things 2026-2

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Instructor

Department of Computer Science

I am a Researcher at INICTEL within the GRFMO research group and an Electronic Engineering graduate from the National University of Engineering (UNI). My work focuses on circuit design, instrumentation, multiphysics simulations, and Small Satellites (CubeSats) in collaboration with APSCO. I am passionate about sharing knowledge and supporting my students, and I look forward to guiding you through this course as we learn together. Outside of academia, I enjoy photography, anime, and indie video games.

Executive Summary

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- **Overview:**

1. Course logistics.
2. About the laboratory course.
3. About the laboratory reports.
4. Groups.
5. Questions and doubts.

Outline

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Course Logistics

About the laboratory course

About the laboratory reports.

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Evaluation

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Teoría 45%	Ciclo regular
Examen Parcial (EP) 20% Examen Final (EF) 25%	Semana 8 Semana 16
Práctica y/o laboratorio 55%	
Laboratorio 1 (L1) 10% Laboratorio 2 (L2) 10% Laboratorio 3 (L3) 10% Laboratorio 4 (L4) 10%	Semana 4 Semana 6 Semana 10 Semana 13
Proyecto (P1) 05% Proyecto (P2) 10%	Semana 7 Semana 16
<i>Nota: P2 no puede ser presentado si no se ha presentado el P1</i>	
100%	

Proyecto se va a empezar a discutir desde las primeras semanas en la sesión de laboratorio.

La primera nota será el P1 que consiste en una presentación de la idea de proyecto, necesaria para poder continuar con el P2.

El P2 se evaluará en formato de una feria de proyecto la última semana de clases.

Los primeros 2 puestos de la feria recibirán automáticamente una bonificación en sus exámenes finales.

Important Rules

- **Zero Tolerance Policy:** Cheating and plagiarism are "very serious" offenses. Sanctions range from a two-semester suspension to permanent expulsion.
- **Assessment Rules:** Unless explicitly stated as a group project, all evaluations are strictly individual. Any unauthorized collaboration or interaction is considered cheating.
- **Prohibited Items:** The use of cell phones, communication apps, or sharing materials (calculators, notes) during exams is forbidden regardless of the reason.
- **Shared Responsibility:** In cases of suspicious similarity between works, both the provider and the receiver of the information will be penalized equally.
- **Disciplinary Process:** Faculty may void exams, issue warnings, and report students to the Department Director. A second warning automatically triggers a formal disciplinary procedure.



Important Rules

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Punctuality:

- **Tolerance** time for arrival to class: **10 minutes.**
- If you **arrive late for the exit test**, you will not be able to take it.
- If you do not arrive for the **Laboratory experience**, you will not be allowed to receive Checkpoints even if your group completes the experience.

Order:

- **No profanity or bad language.**
- **Respectful treatment.**

Questions?

Call with **respect** (“Profesor”)

Logistics

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Official communications, such as excused absences, technical issues, or matters of similar importance, must be sent solely and exclusively via email to *garias@utec.edu.pe*

Case: One or more team members go AWOL.

Minor inquiries or quick questions may also be sent via Gmail messages (Hangouts/Chat)

Regarding the Laboratory Reports

Turnitin Originality will be used. If your laboratory report shows a **plagiarism percentage higher than 25%, 3 points will be automatically deducted with no right to appeal.**

Each report must include a list of the people who worked on the project/lab and their specific contributions, e.g.:

UNIVERSIDAD DE INGENIERÍA Y TECNOLOGÍA
Computer Science

Título del Reporte
Curso

Integrantes del Grupo

- Integrante 1: revisión bibliográfica/búsqueda de normas/
- Integrante 2: implementación de nodo sensor/revisión bibliográfica
- Integrante 3: implementación de nodo sensor/revisión bibliográfica/frontend

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- Introduces fundamental concepts of IoT paradigm for design and implementation.
- **Objectives:**
 - Show the principles of interconnected IoT systems.
 - Explain the interaction between computing and sensing for IoT.
 - Implement an IoT application system as a solution.
- **Content:**

Distributed in two modules:

 - First module, from Week 1 to Week 8
 - Second module, from Week 9 to Week 16

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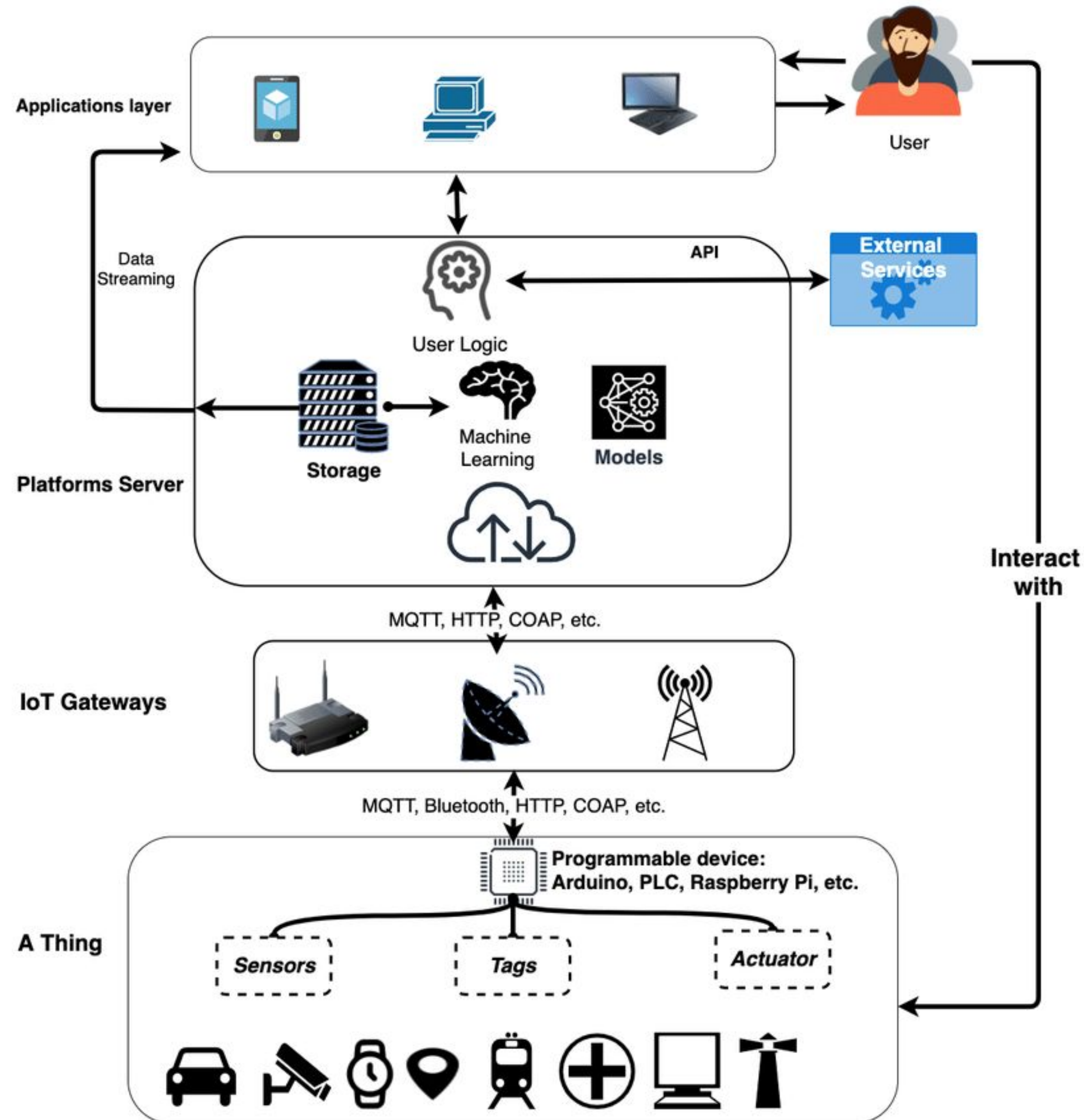
- The laboratories will be carried out with **electronic components kits**.
- Each group will be **assigned a lab kit**, which have to be taken care of by the students themselves during the sessions. At the end of the session, **the kits must be returned**.

So, what are we going to study along this half-year?

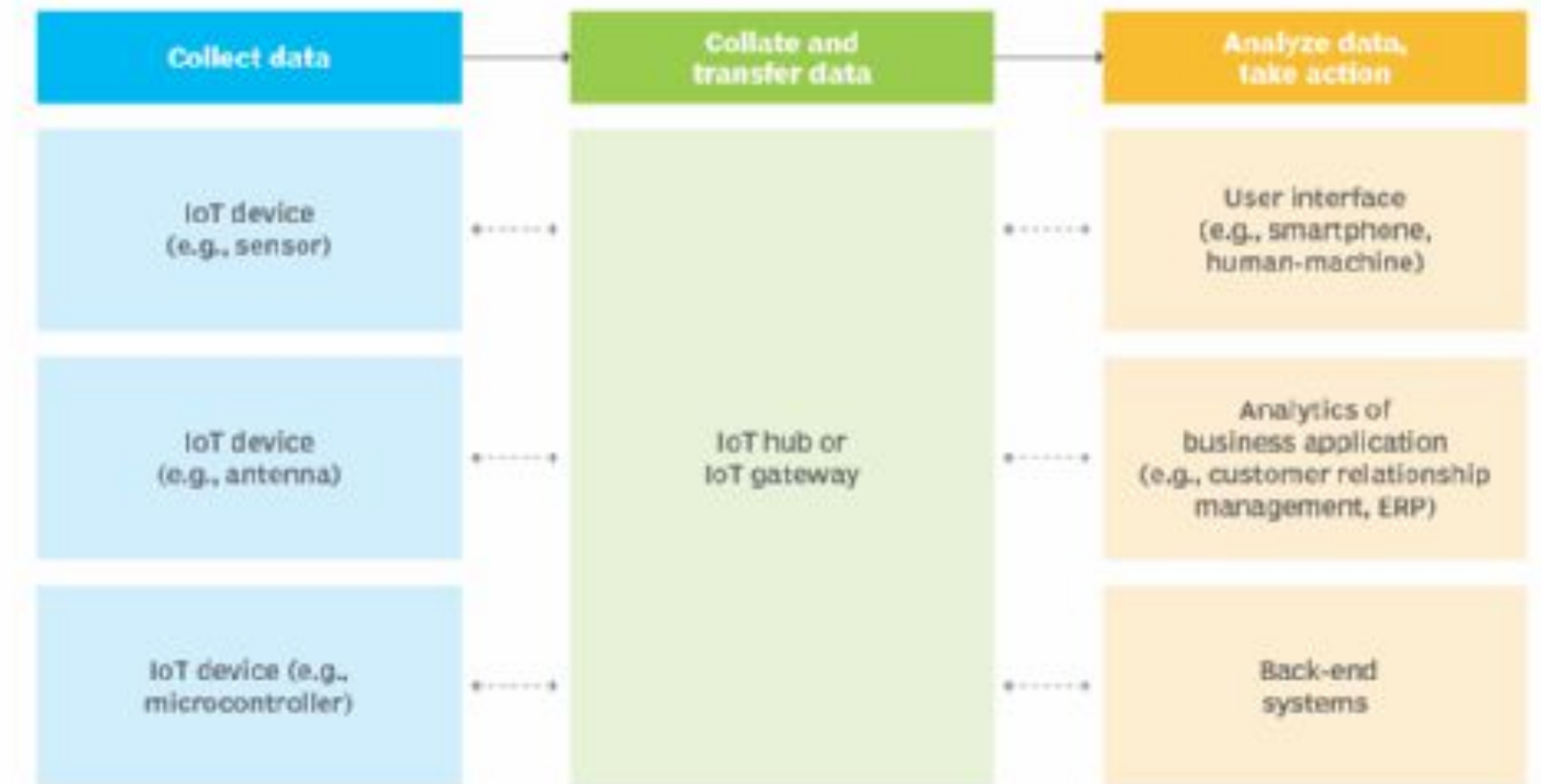


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Example of an IoT system

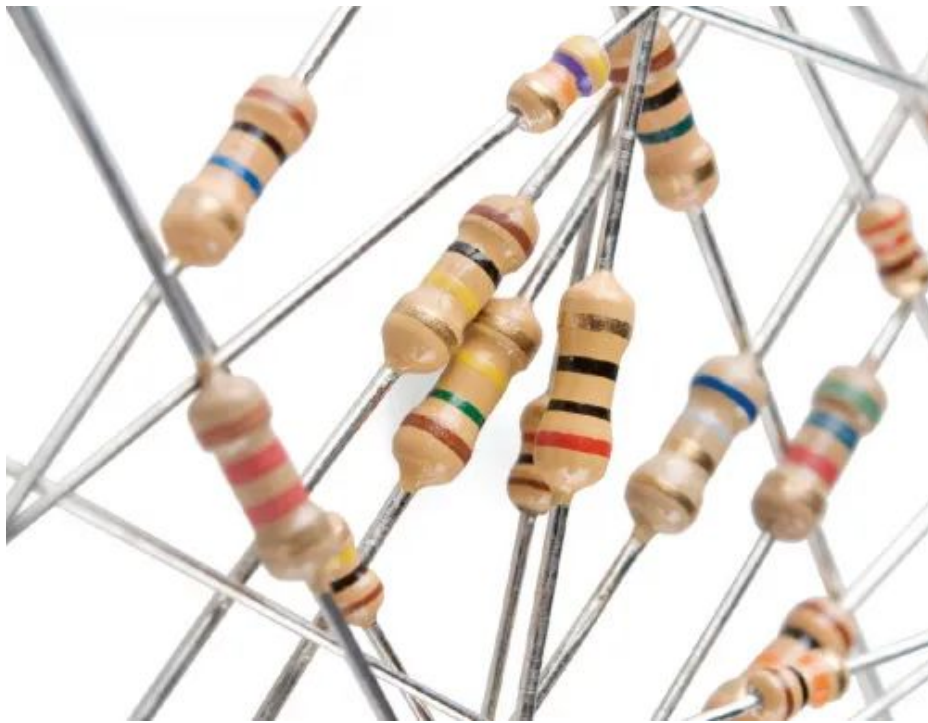


Source: TechTarget

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- Electronic Components



Resistencias



Diodos LED



Capacitores



Diodos



Botones



Inductores

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- SMD Components



SMD Capacitors



SMD connectors



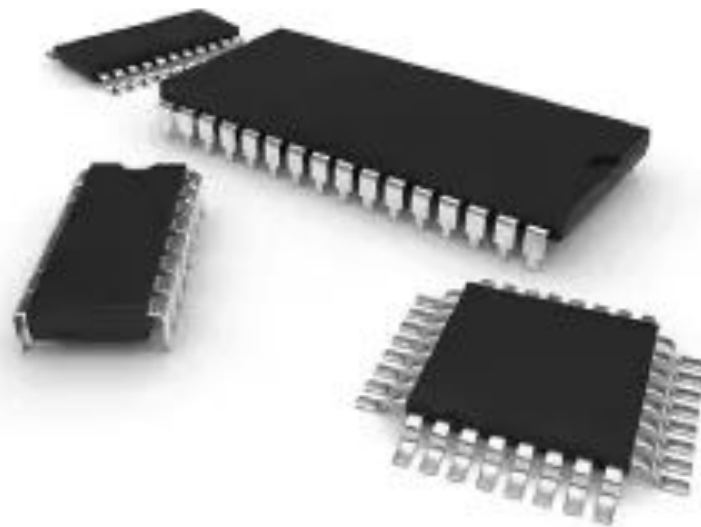
SMD Crystal Oscillators



SMD Diodes



SMD Inductors



SMD Integrated Circuits (ICs)



SMD Resistors



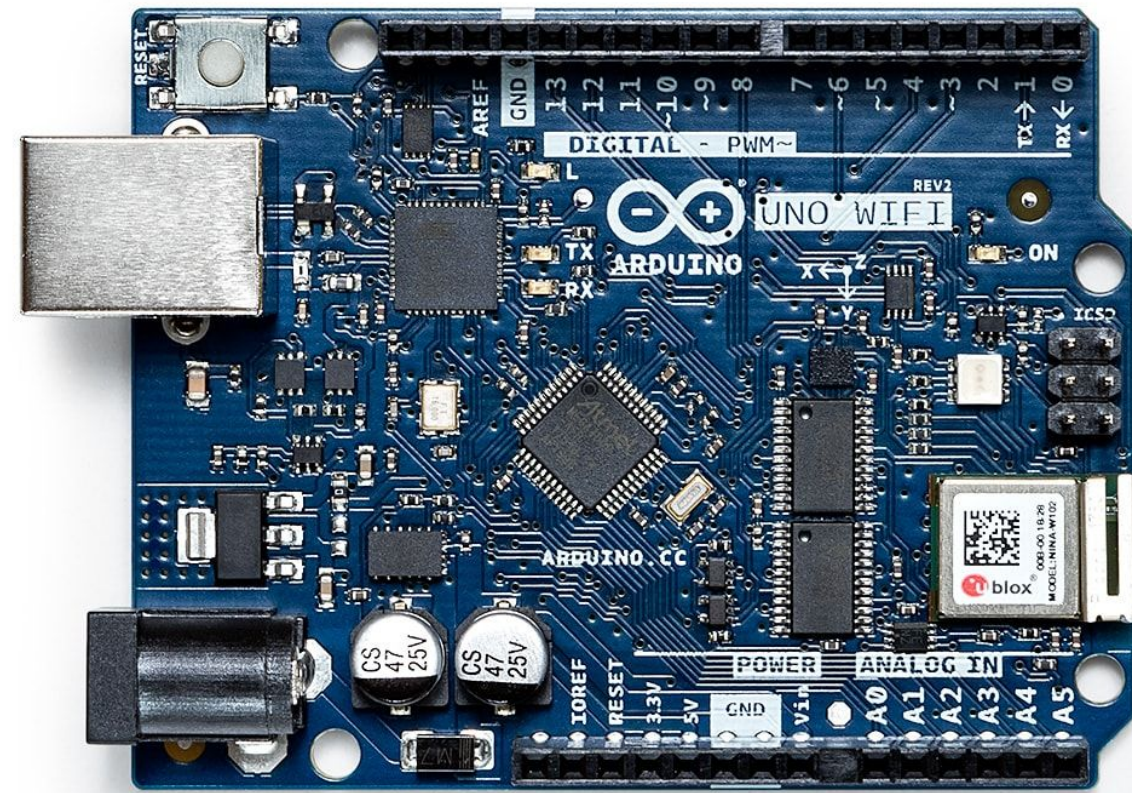
SMD Switches



comparison	Metric code	Imperial code	comparison
0.1x0.1 mm	0402	01005	0.01x0.01 in (10x10 mils)
	0603	0201	
	1005	0402	
	1608	0603	
1x1mm	2012	0805	0.1x0.1 in (100x100 mils)
	2520	1008	
	3216	1206	
	3225	1210	
	4516	1806	
	4532	1812	
	5025	2010	
1x1 cm	6332	2512	0.5x0.5 in (500x500 mils)
	Actual size		

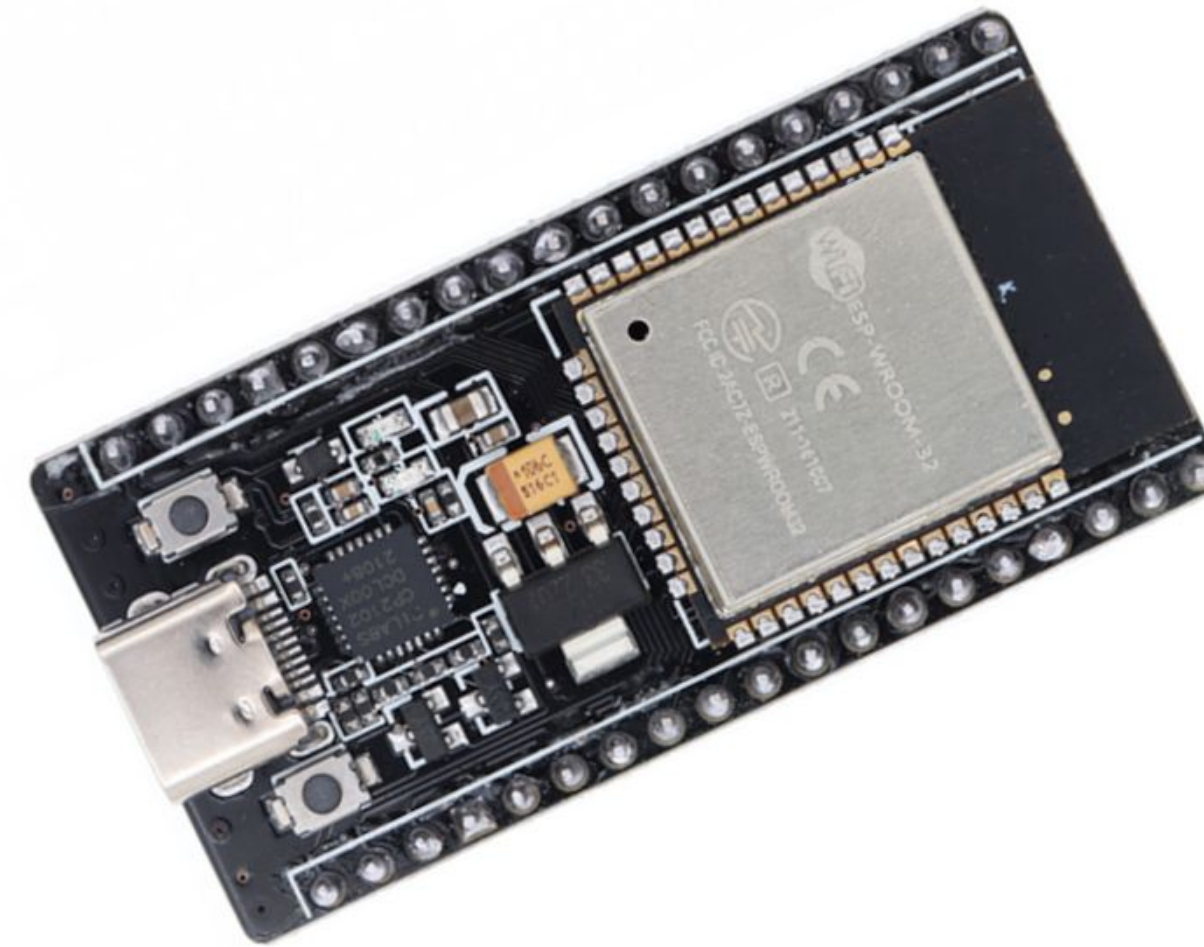
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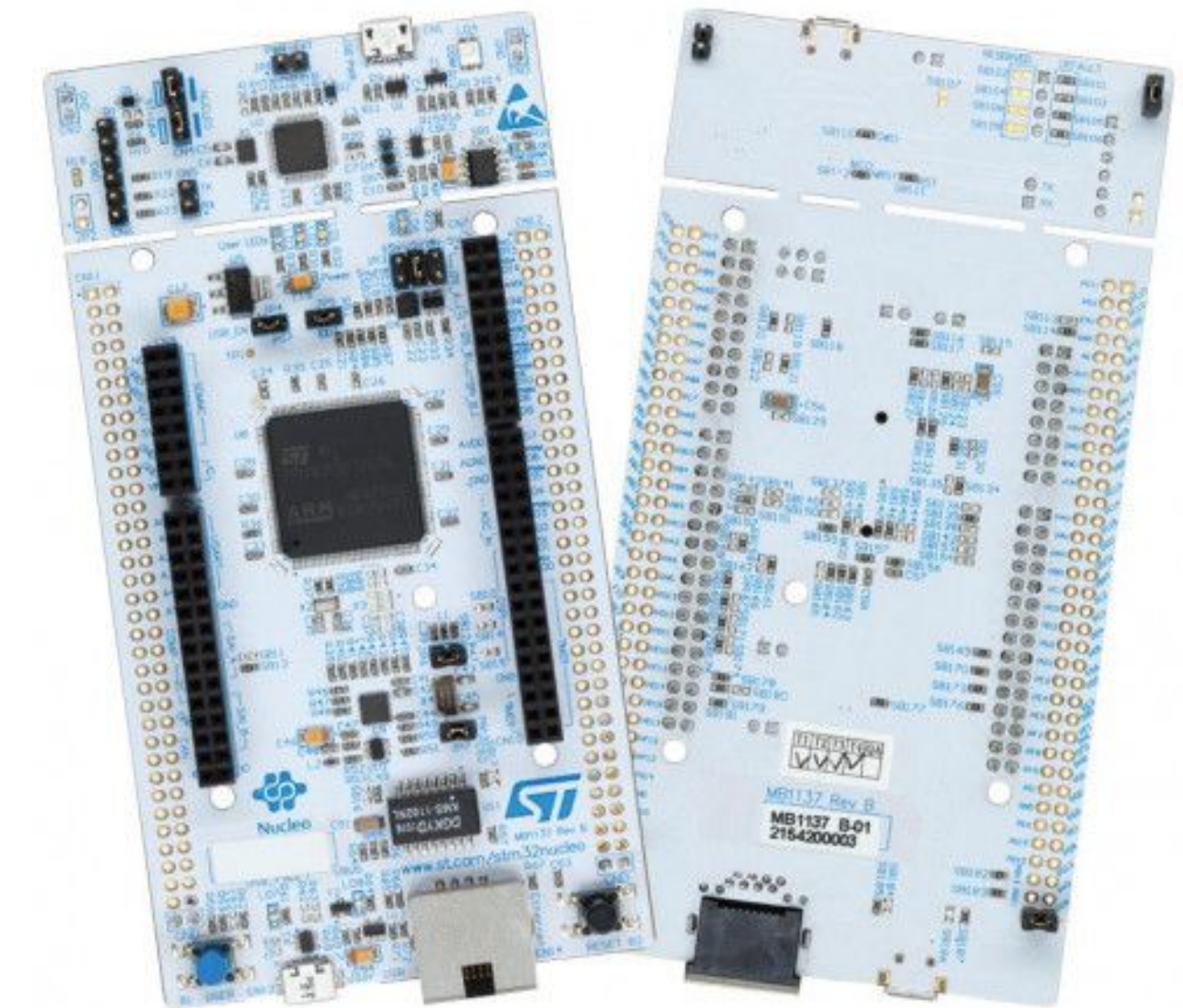
Arduino UNO

Arduino MEGA



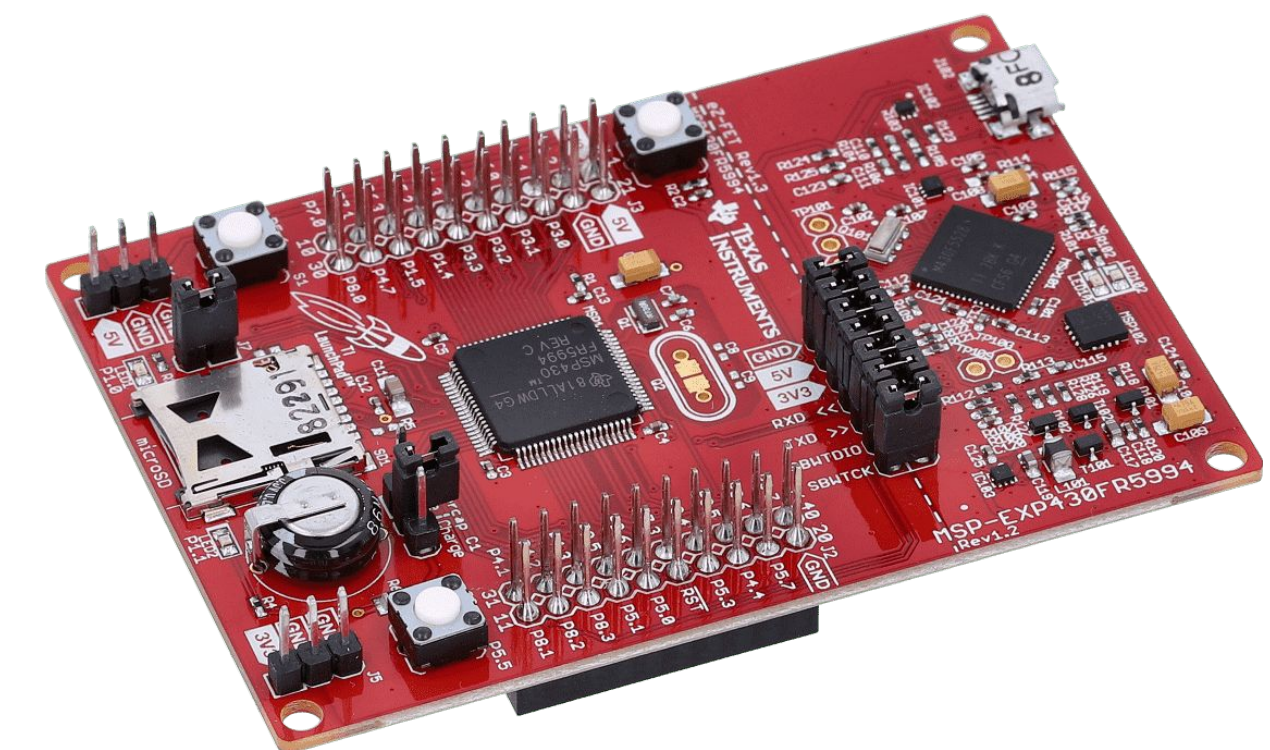
ESP32

Raspberry Pi



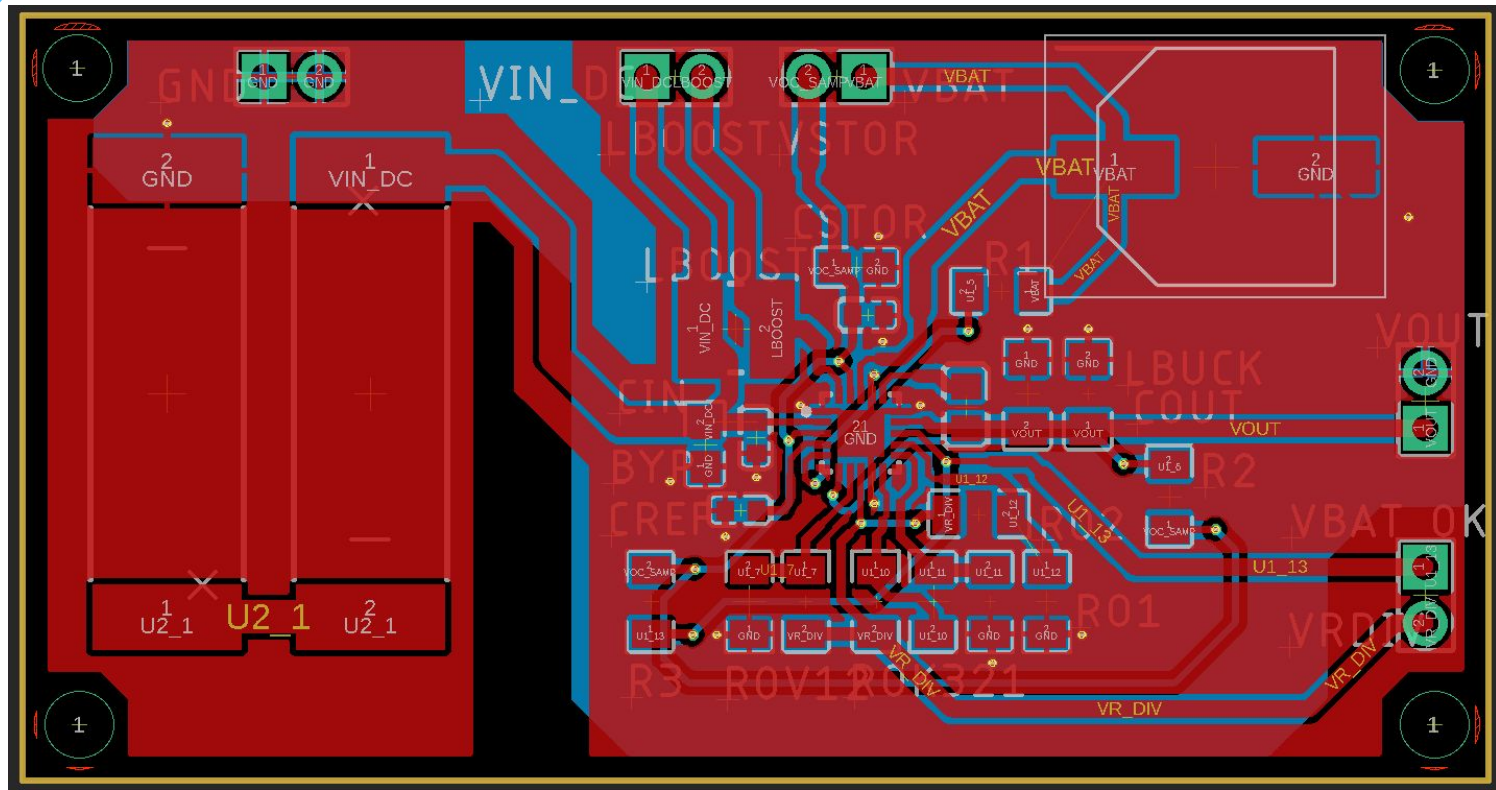
STM32 Nucleo 144

MSP430

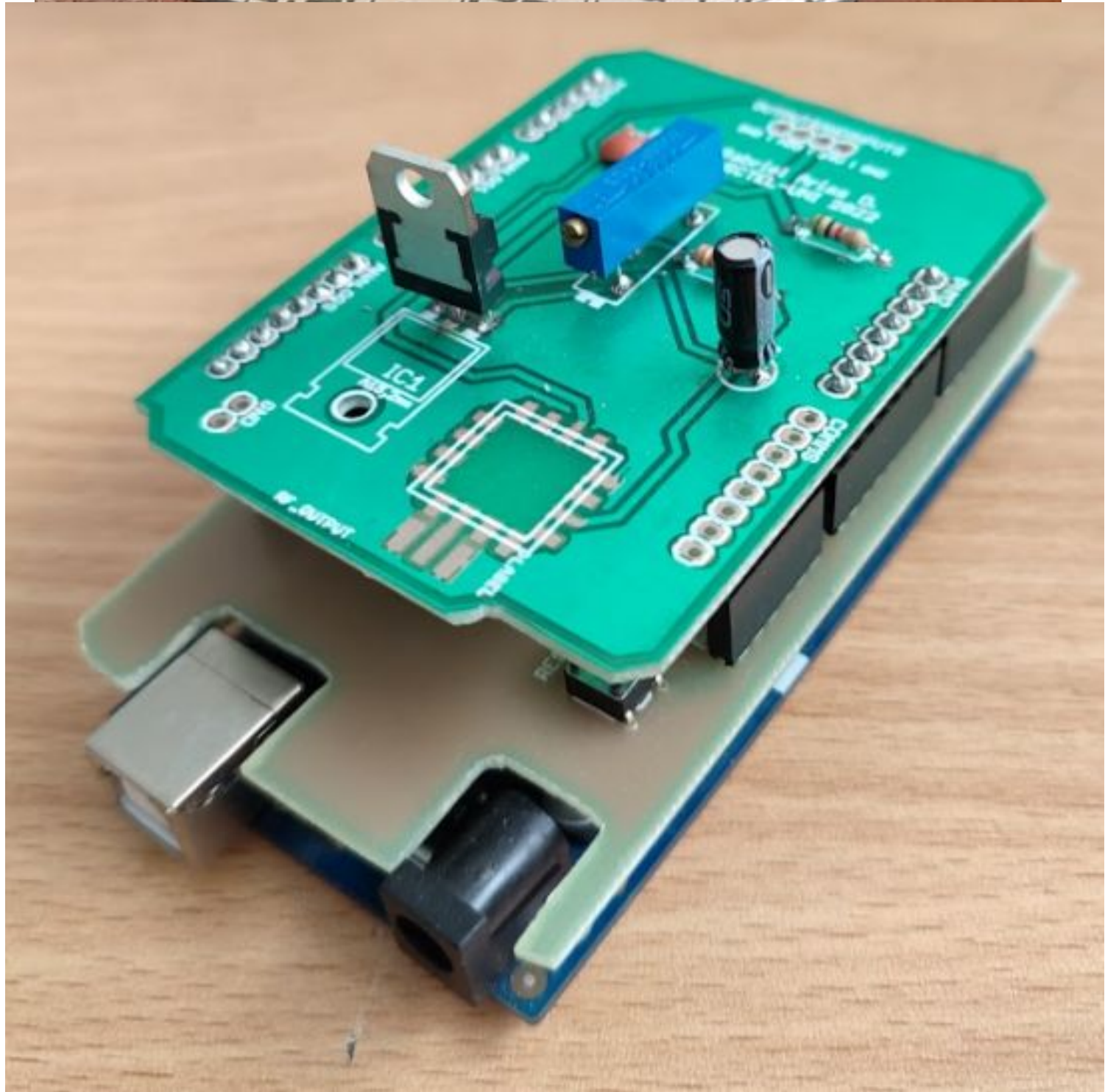
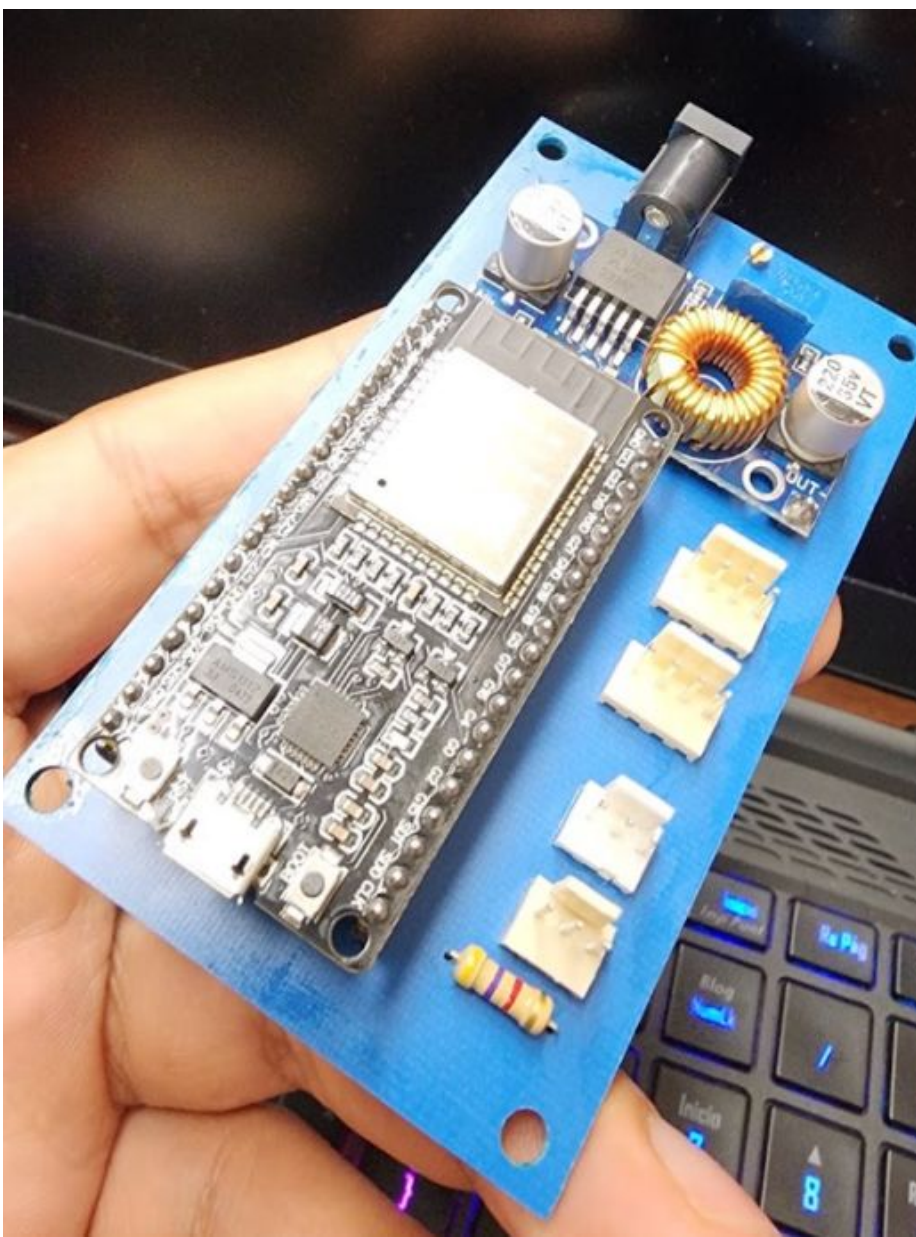
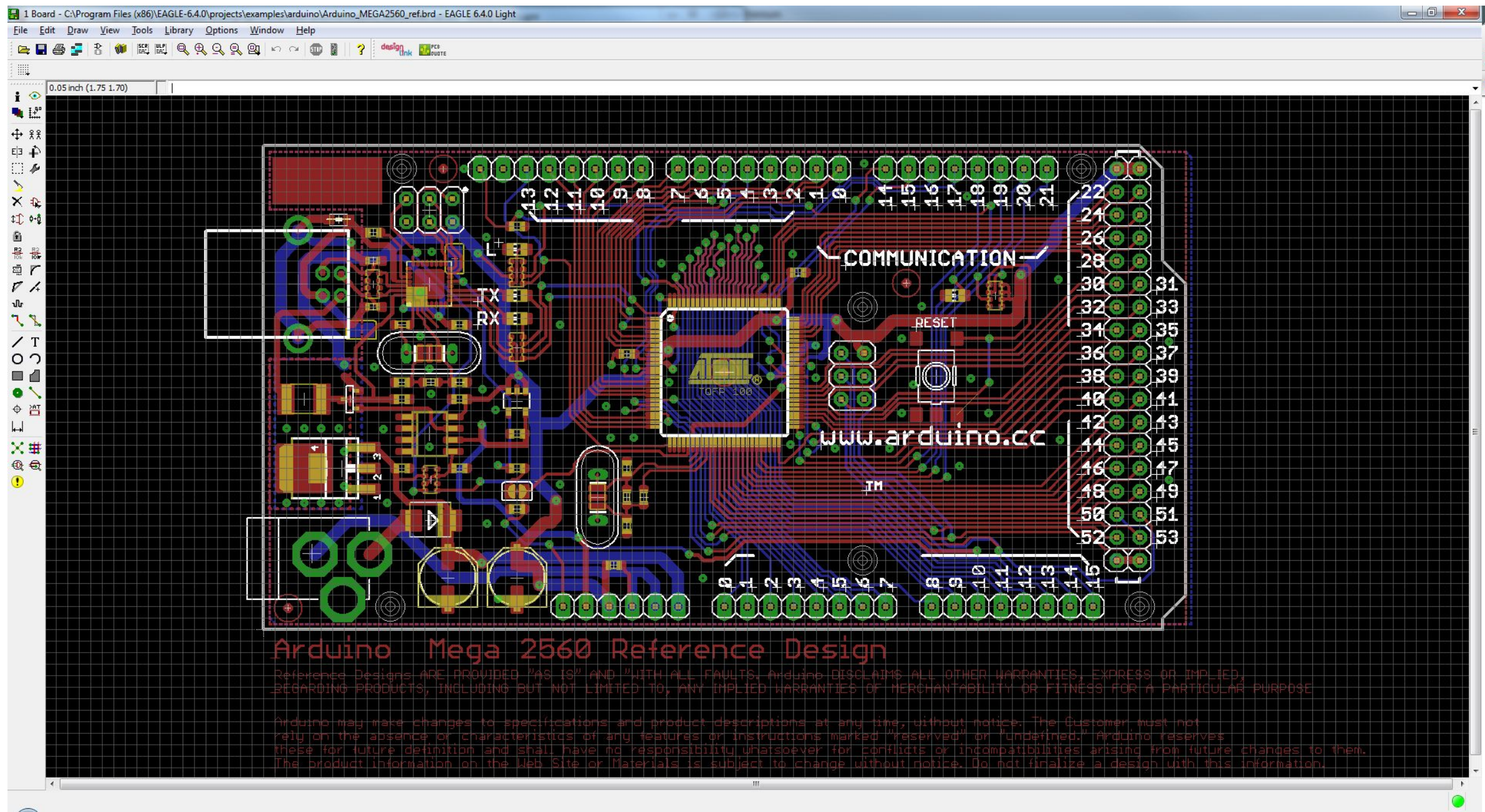
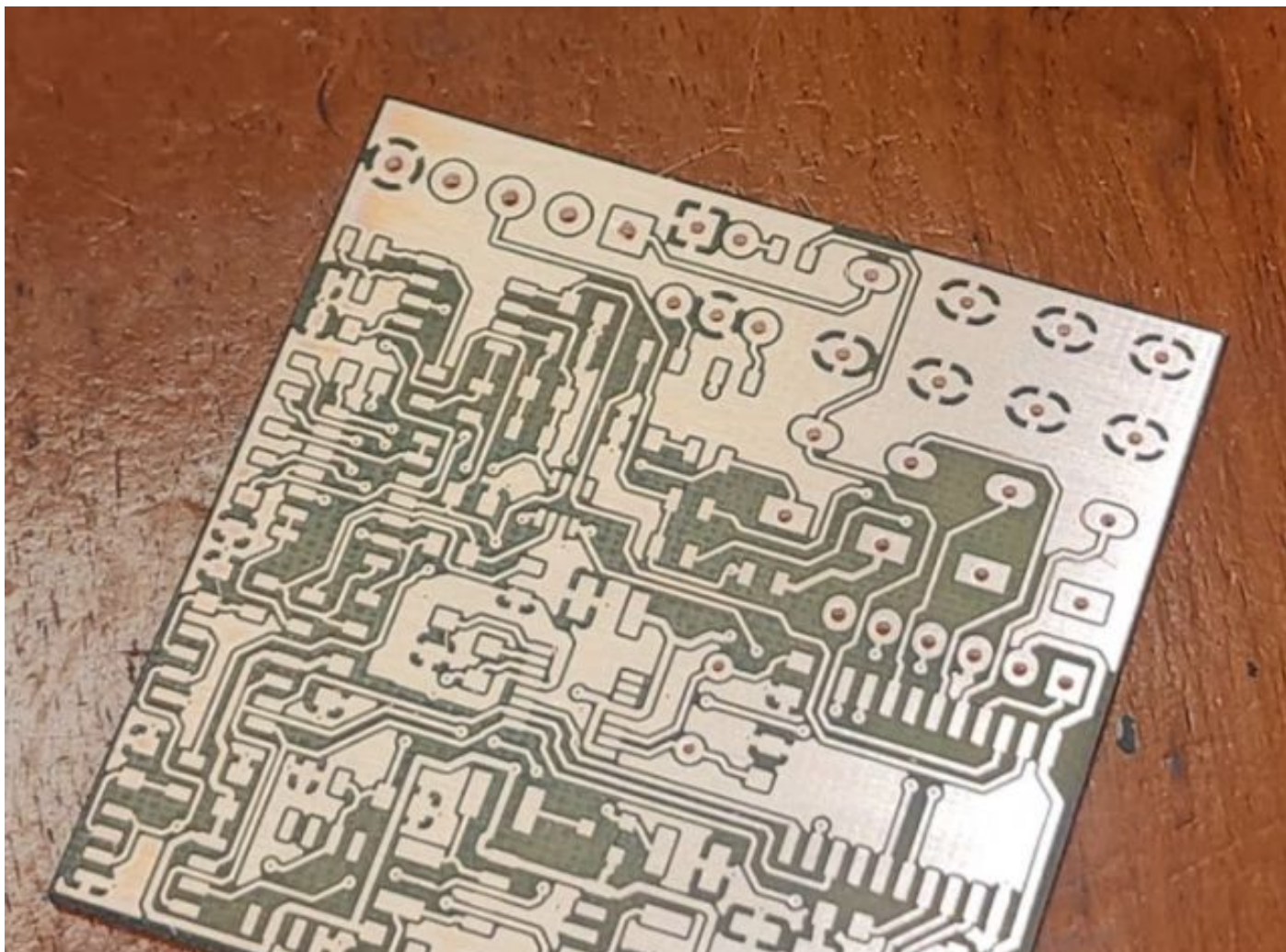
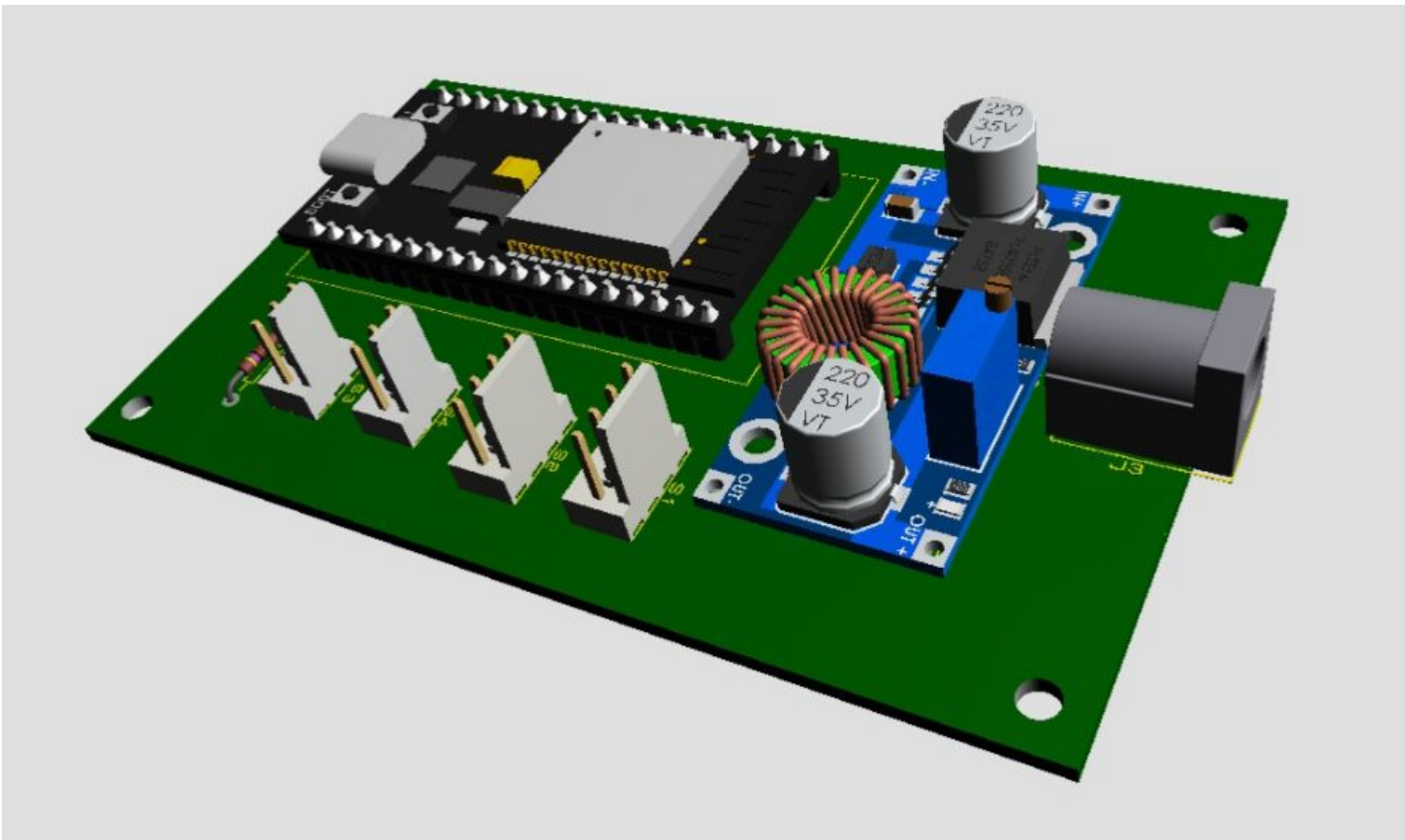


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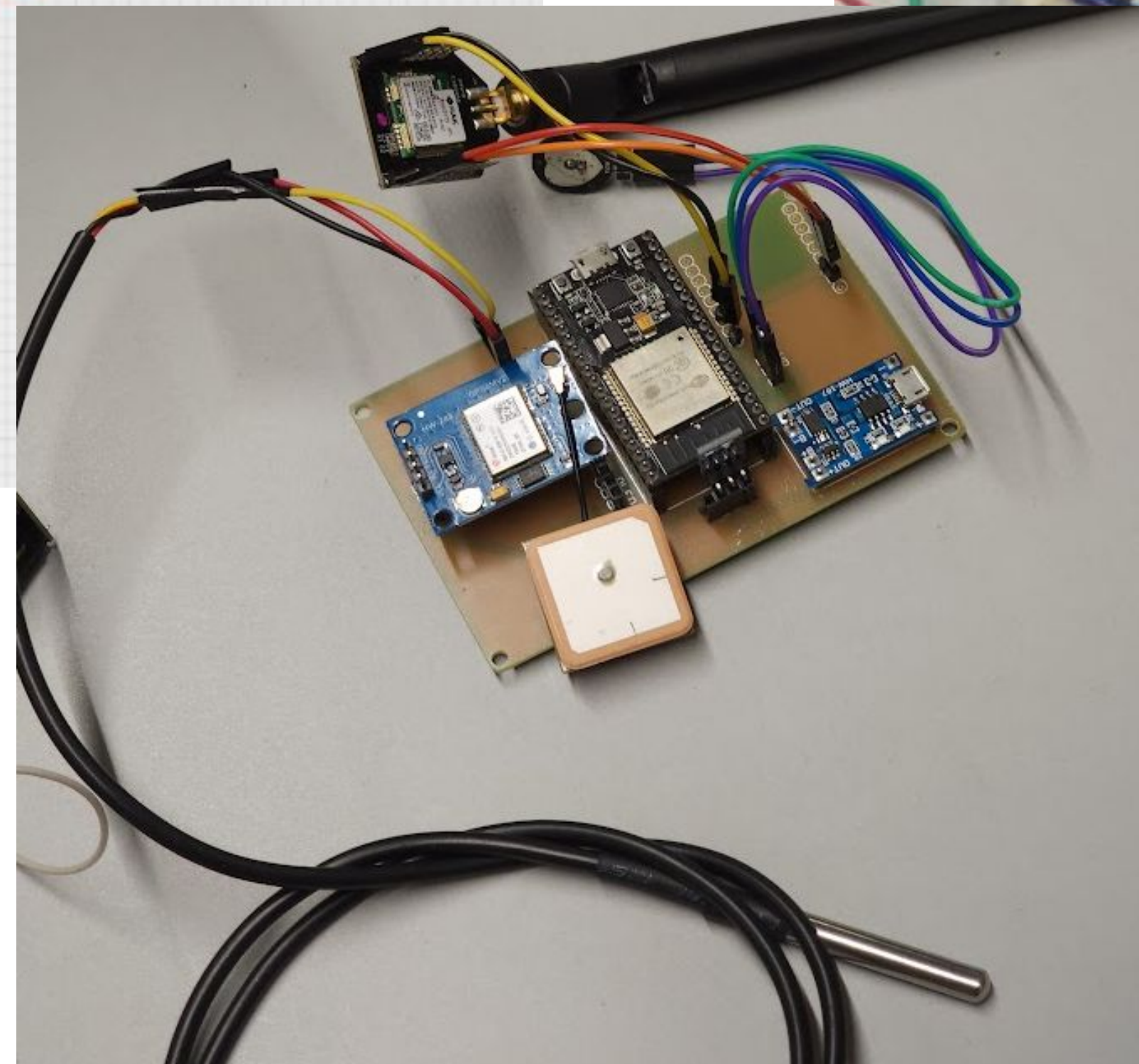
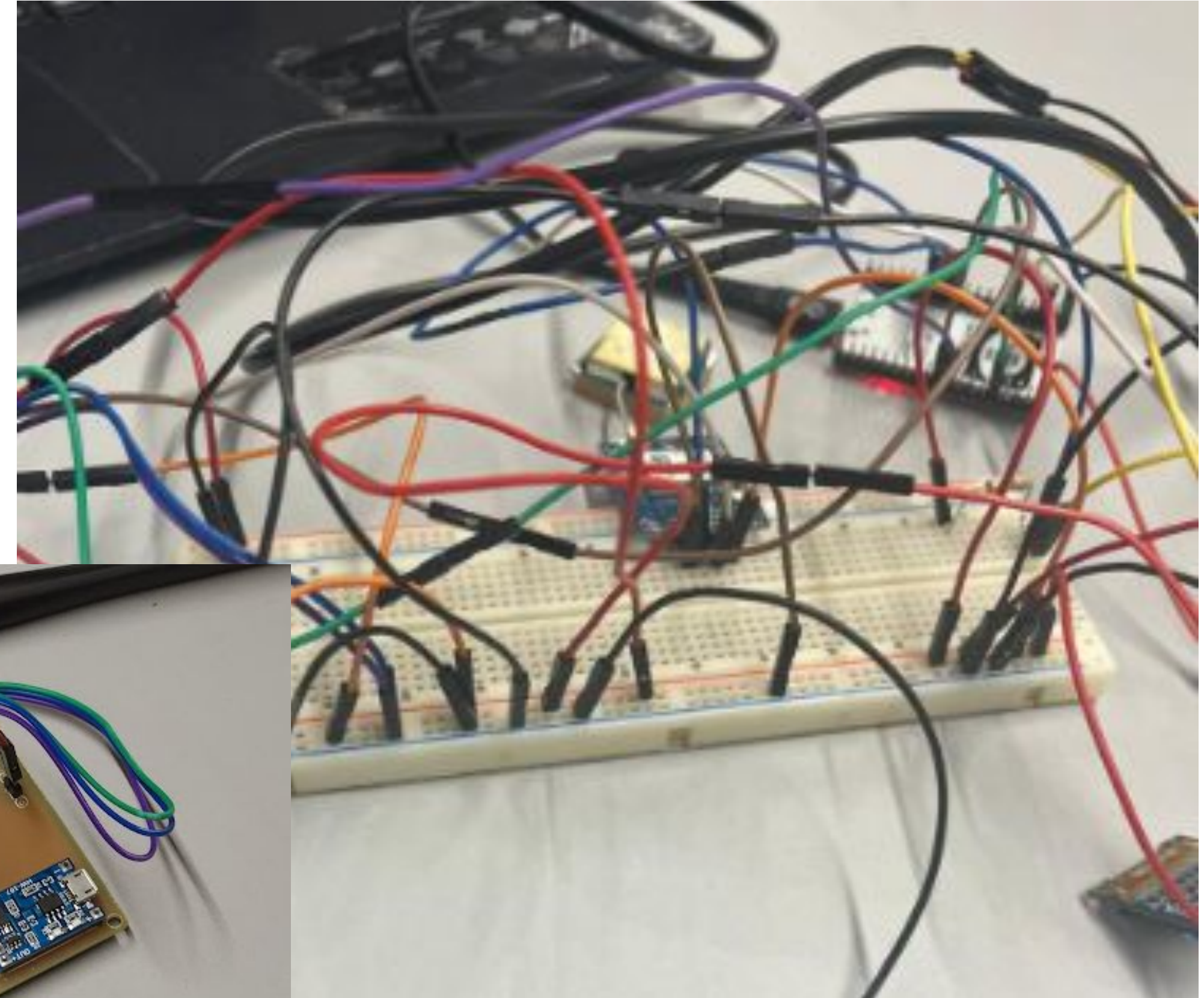
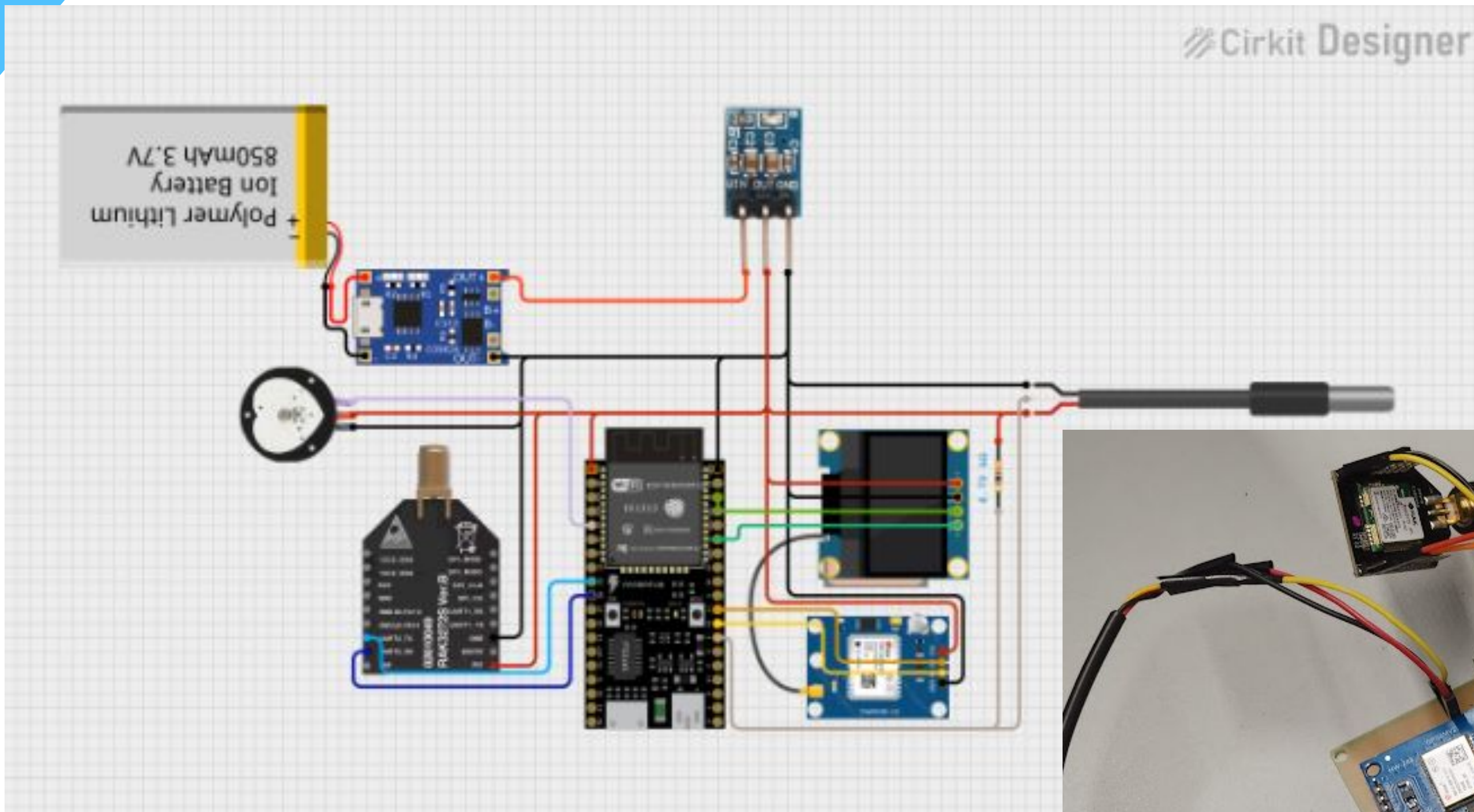


Eagle Circuit Design



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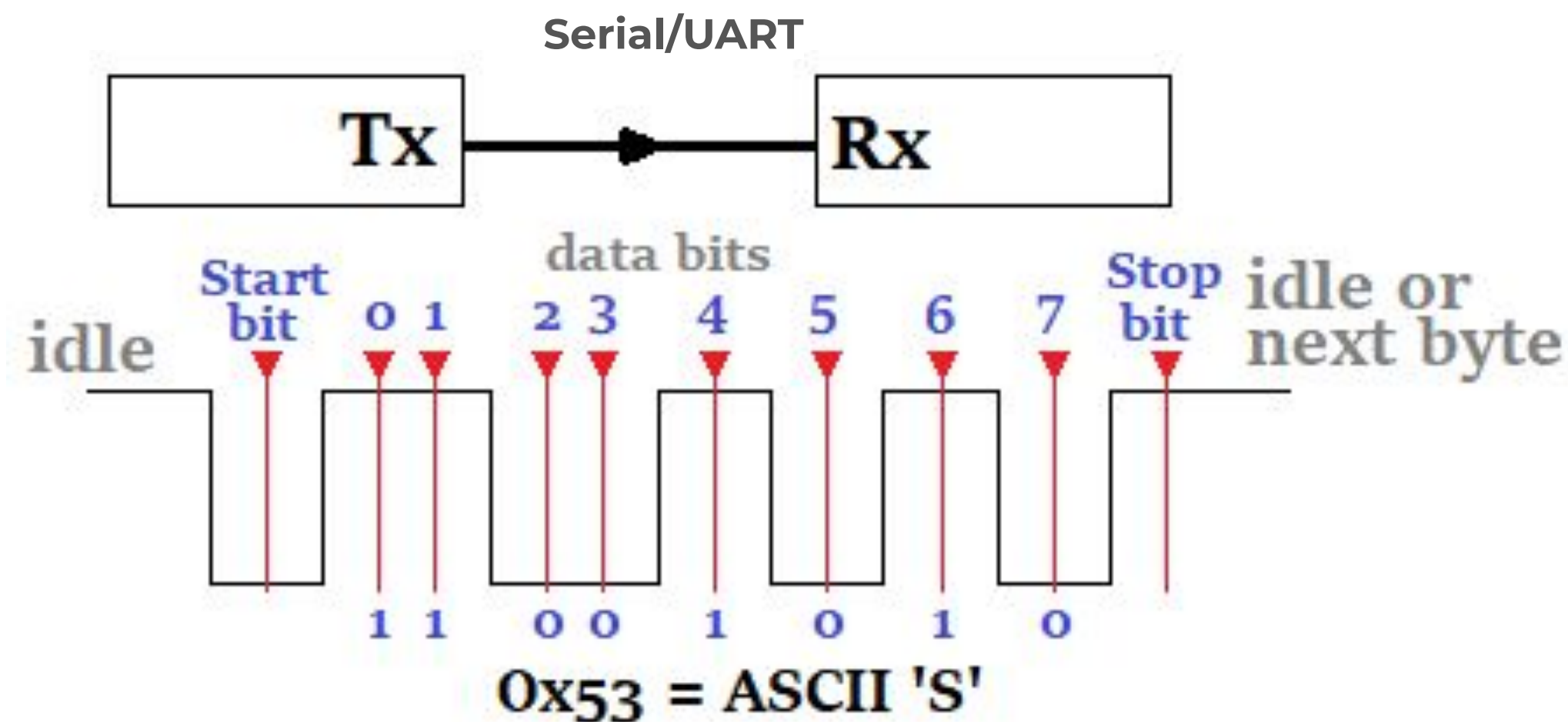
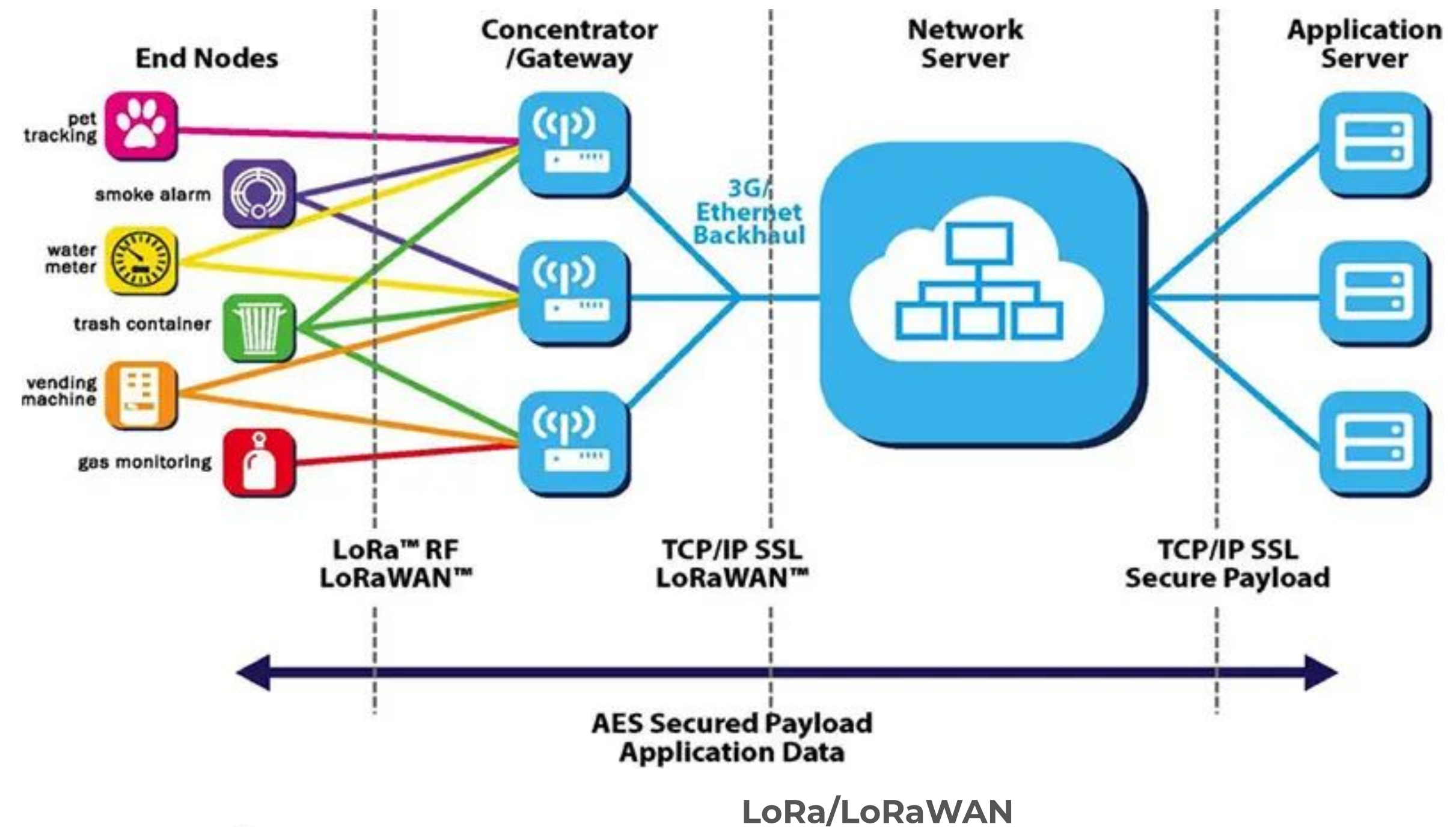
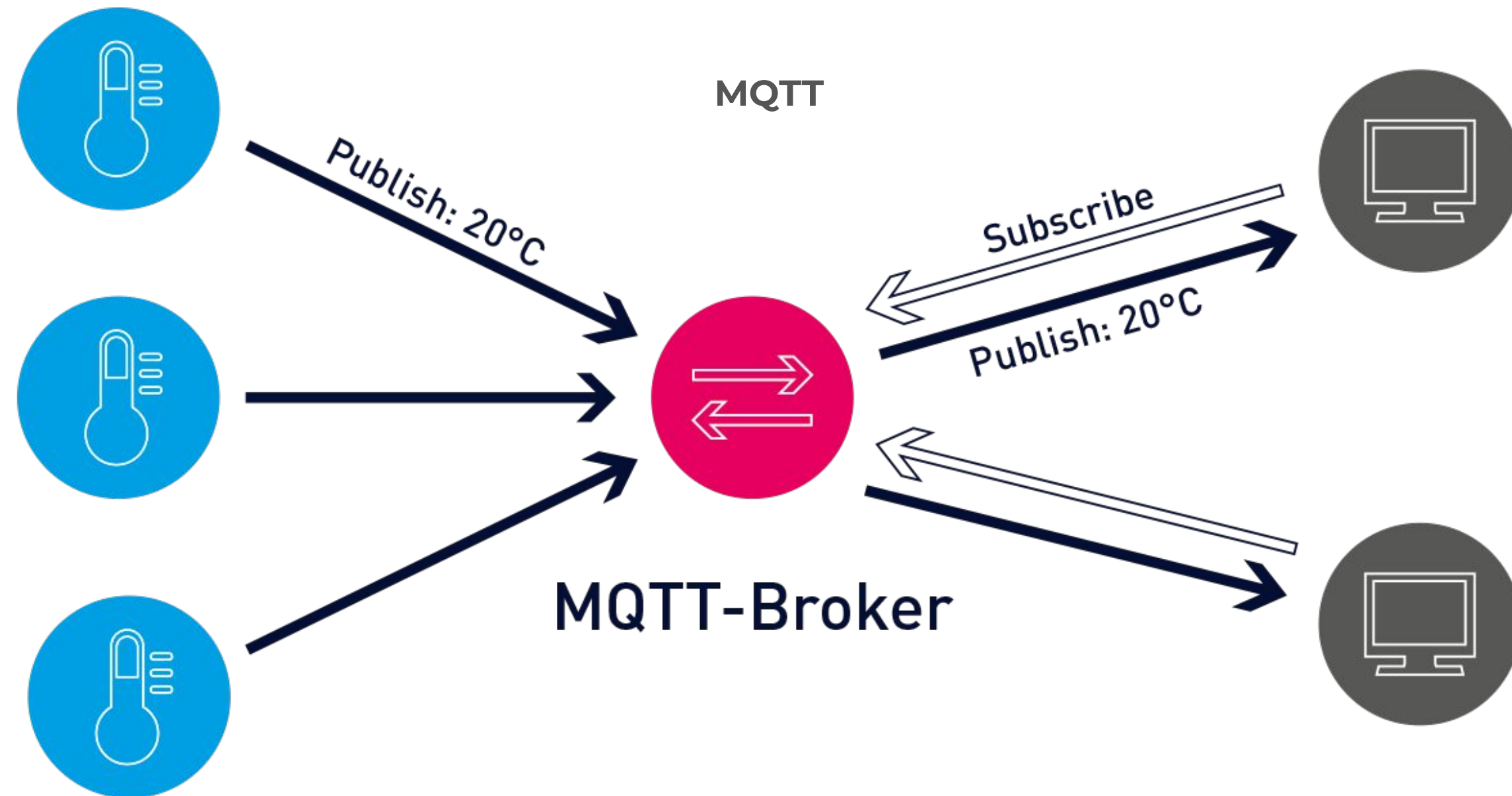
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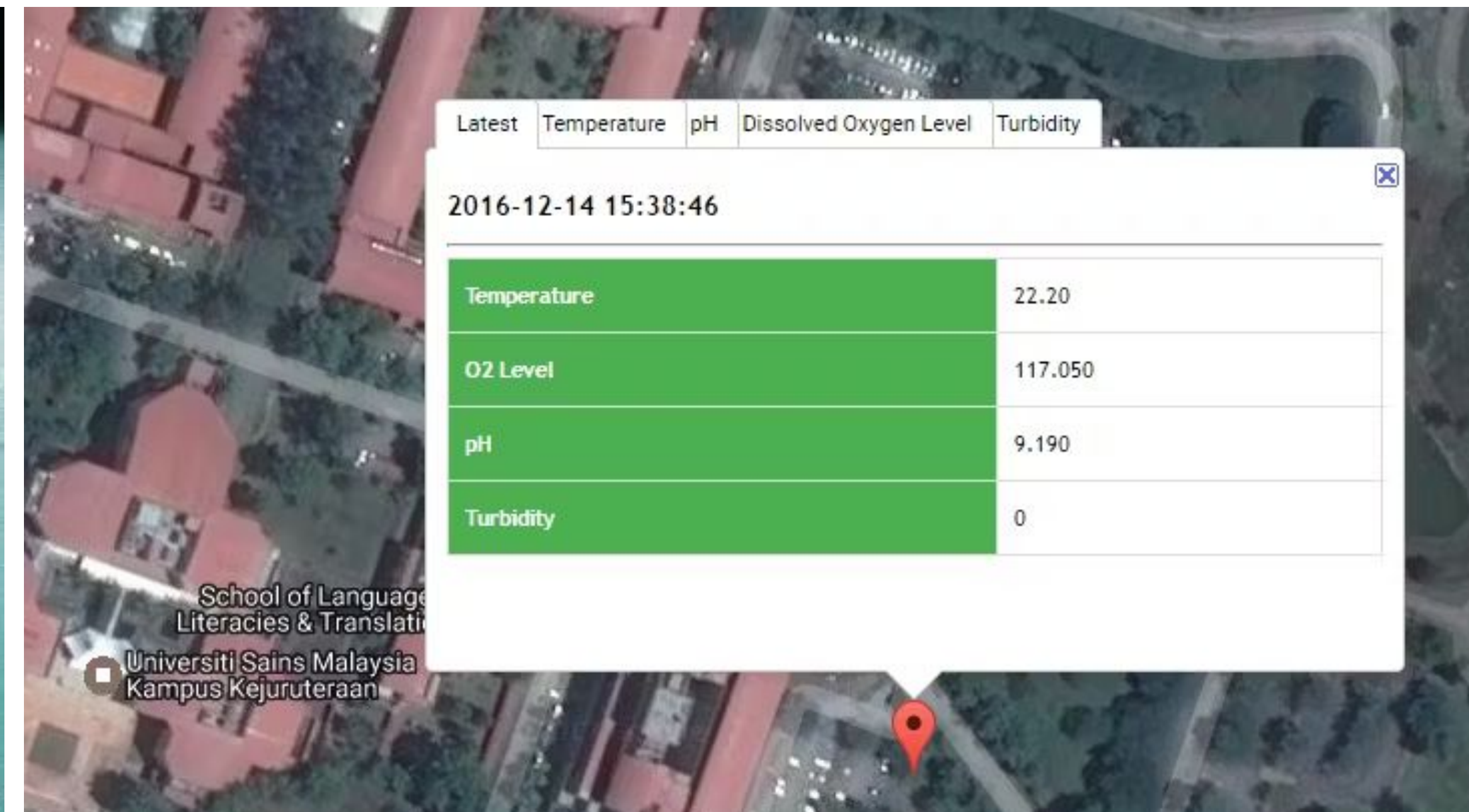
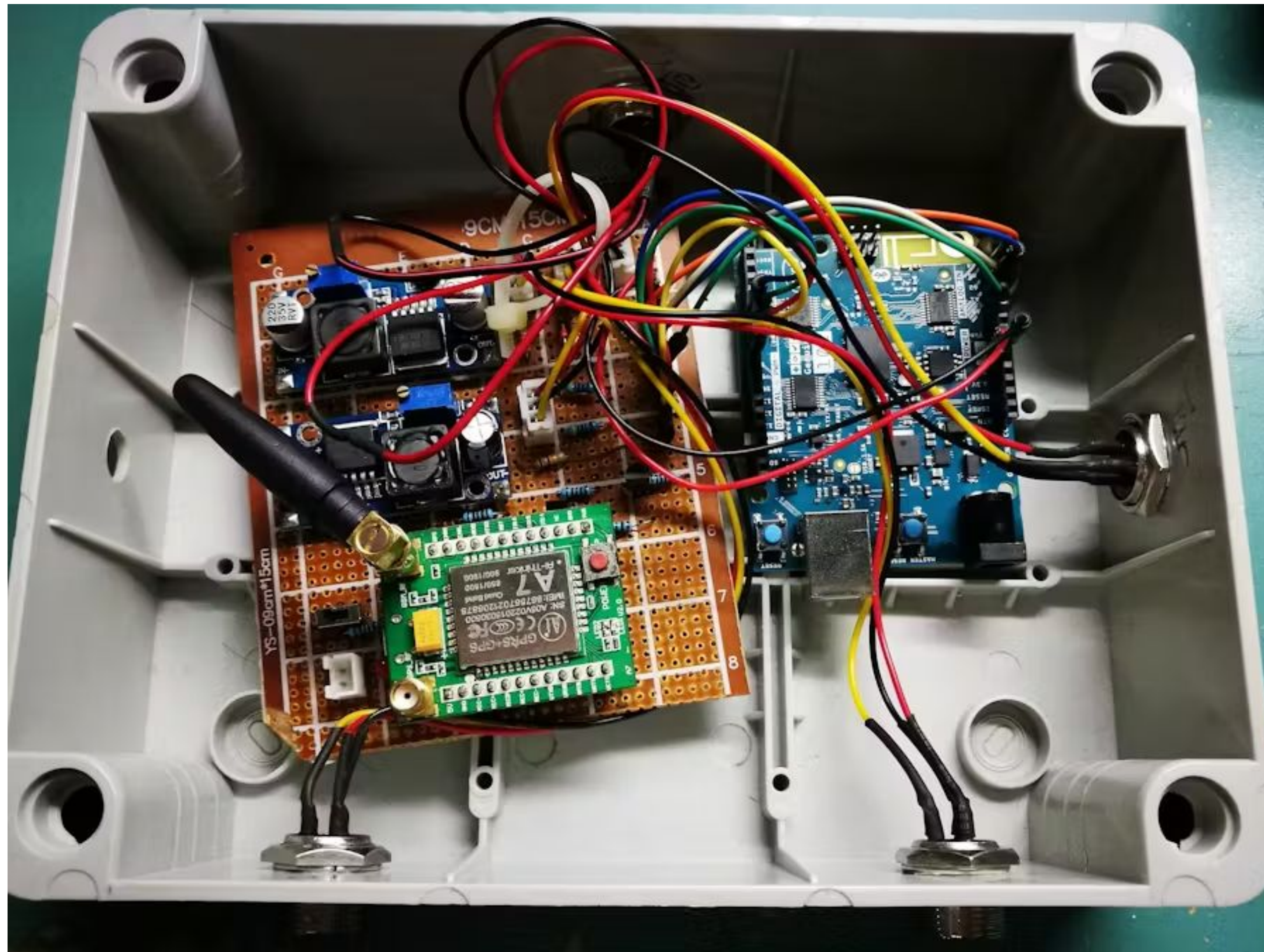
Communication Protocols



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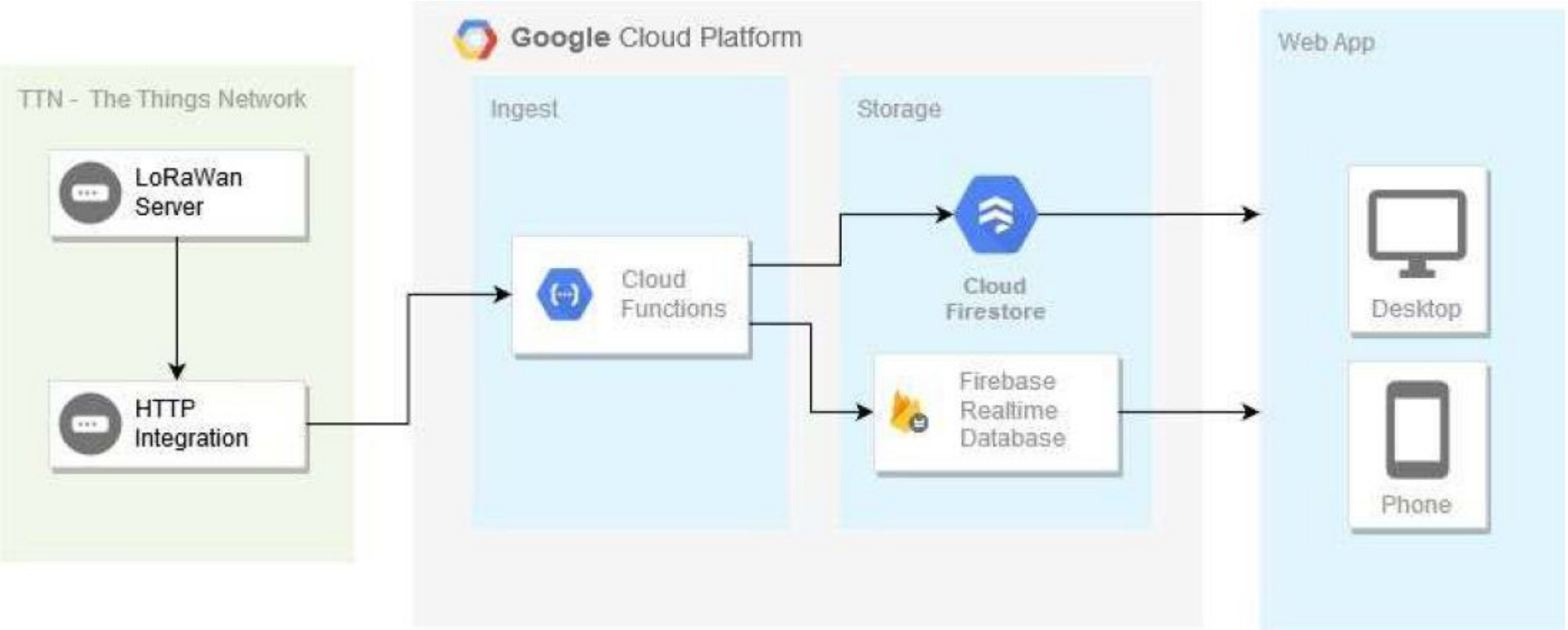
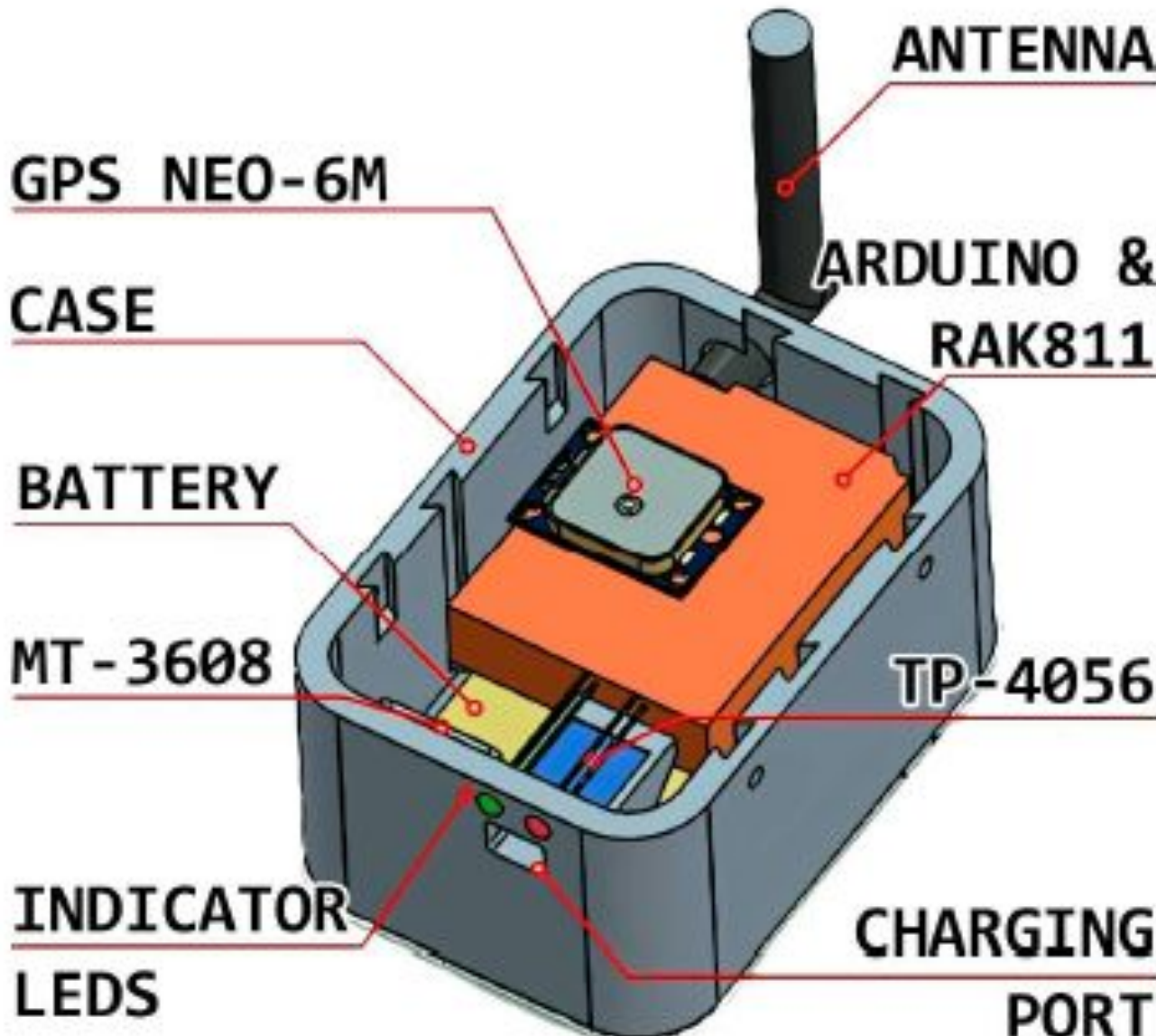
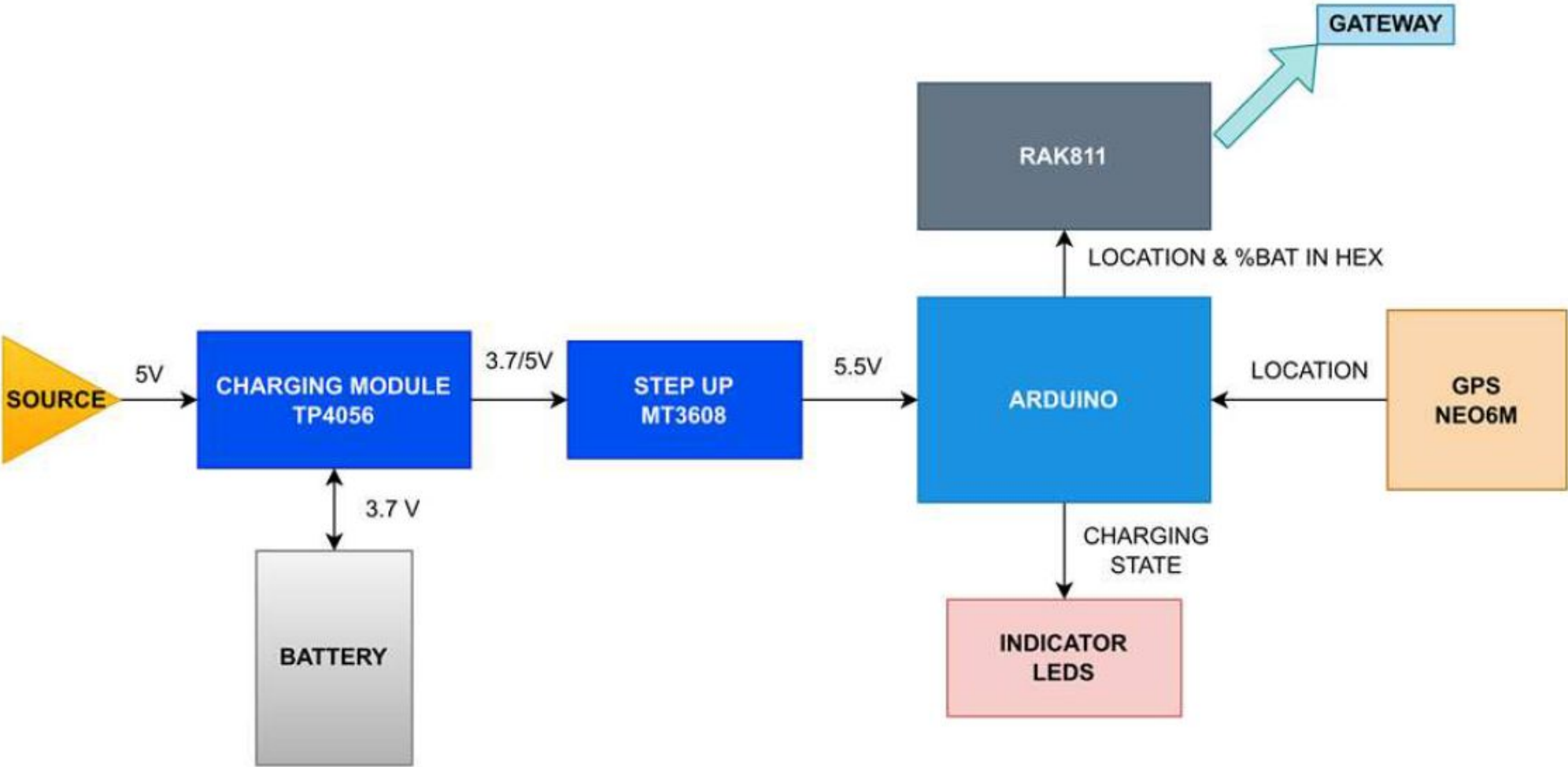
You are going to become an experts!



Source: Chan, Hao J.

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Laboratorios:

- **LAB 0.** Resolución de **Circuitos Eléctricos y Simulación**
- **LAB 1.** Introducción a **Arduino**
- **LAB 2.** **Sensores y Actuadores**
- **LAB 3.** Protocolos de Comunicación: **MQTT**
- **LAB 4.** Protocolos de Comunicación **LPWAN: LoRa**

Workshop:

- **W1.** Diseño de Placas de Circuito Impreso (**PCB**) con KiCad

Outline

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Questions and doubts.

Laboratory Reports

The reports **must** be written in LaTeX (Overleaf).

- **Recommended template*:**

<https://www.overleaf.com/latex/templates/ieee-journal-paper-template/jbbbdkztxrd>

The submission should consist of:

1. Introduction
 - a. Theoretical Framework
 - b. State of the Art
2. Development of the experience (including discussion)
3. Conclusions
4. References

* Another template with one or two columns is allowed.



Laboratory Reports

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- It is a **requirement** to have checkpoints marked by the teacher, so that the development of your lab report can be reviewed.

Without checkpoints, the developed section is not reviewed.

- The **deadline** is defined by the teacher.
- Submission of checkpoints is **only given during lab time.**

No email validation is allowed.

Grading

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- Laboratory Report: 12 pts.
- Exit Test: 8 pts.

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- All laboratory experiments will be done in **groups of 3 students.**
- The groups are **fixed**, *no changes* will be made between labs.
- In case a student is not present without prior notice or authorization from the teacher, **the checkpoint will not be counted for that student (even if the group has done it).**

Project!

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The Final Project has **two main evaluation stages**:

P1 – Project Profile (Week 08)

- Proposal presentation (slides + report).
- Includes introduction, problem statement, theoretical framework, state of the art, methodology, components list, microcontroller, block diagram, objectives & scope.

P2 – Final Presentation (Week 16)

- Oral presentation + prototype demonstration.
- Advertising-style **video** required*.
- Final report in **LaTeX**.

Approval: Every group must have their project validated by Week 4-5.

Key Deliverables: Functional IoT prototype, implementation evidence, results analysis, market description, conclusions, and references. Also a proper housing/case and/or PCB.

Video: We are making a contest of the best advertising video for bonus points!

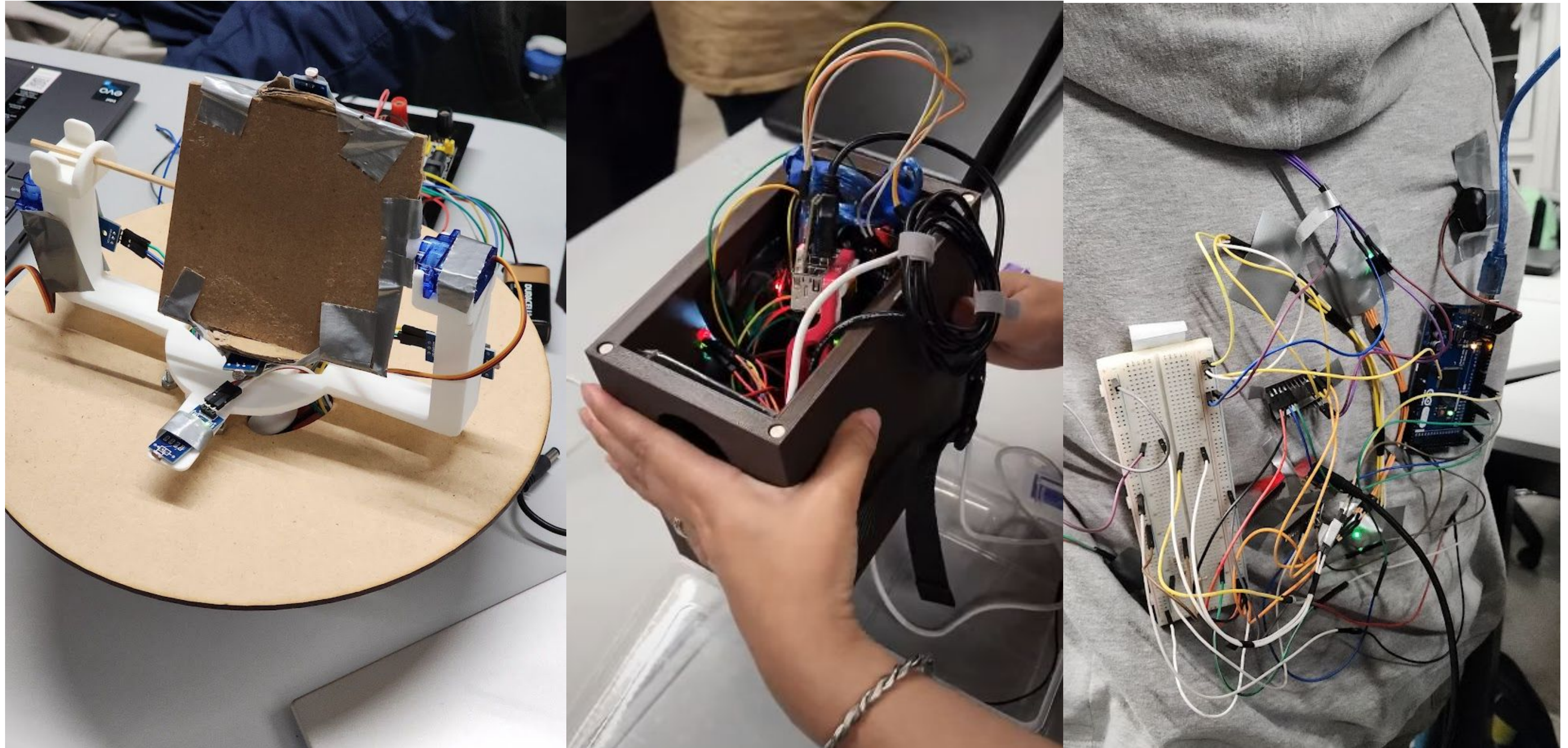
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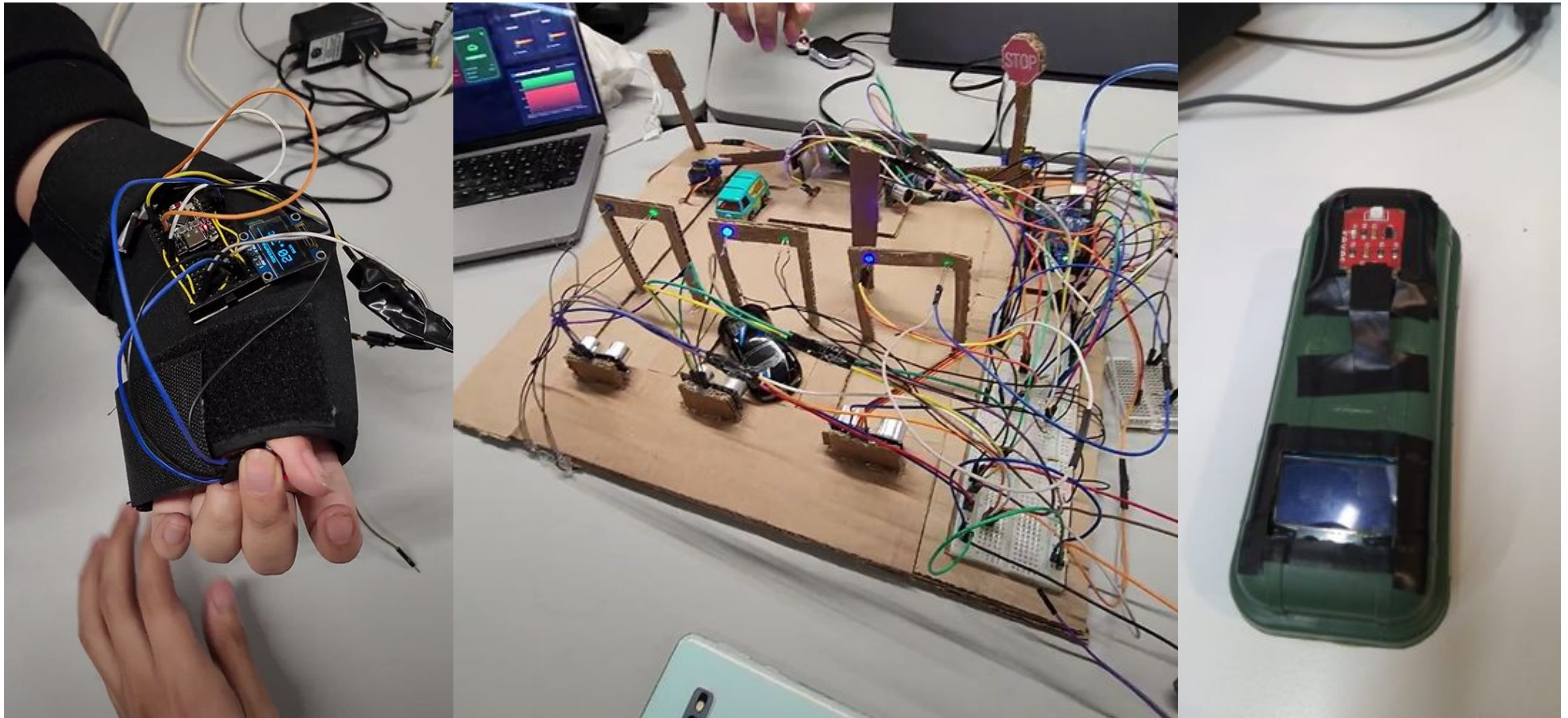
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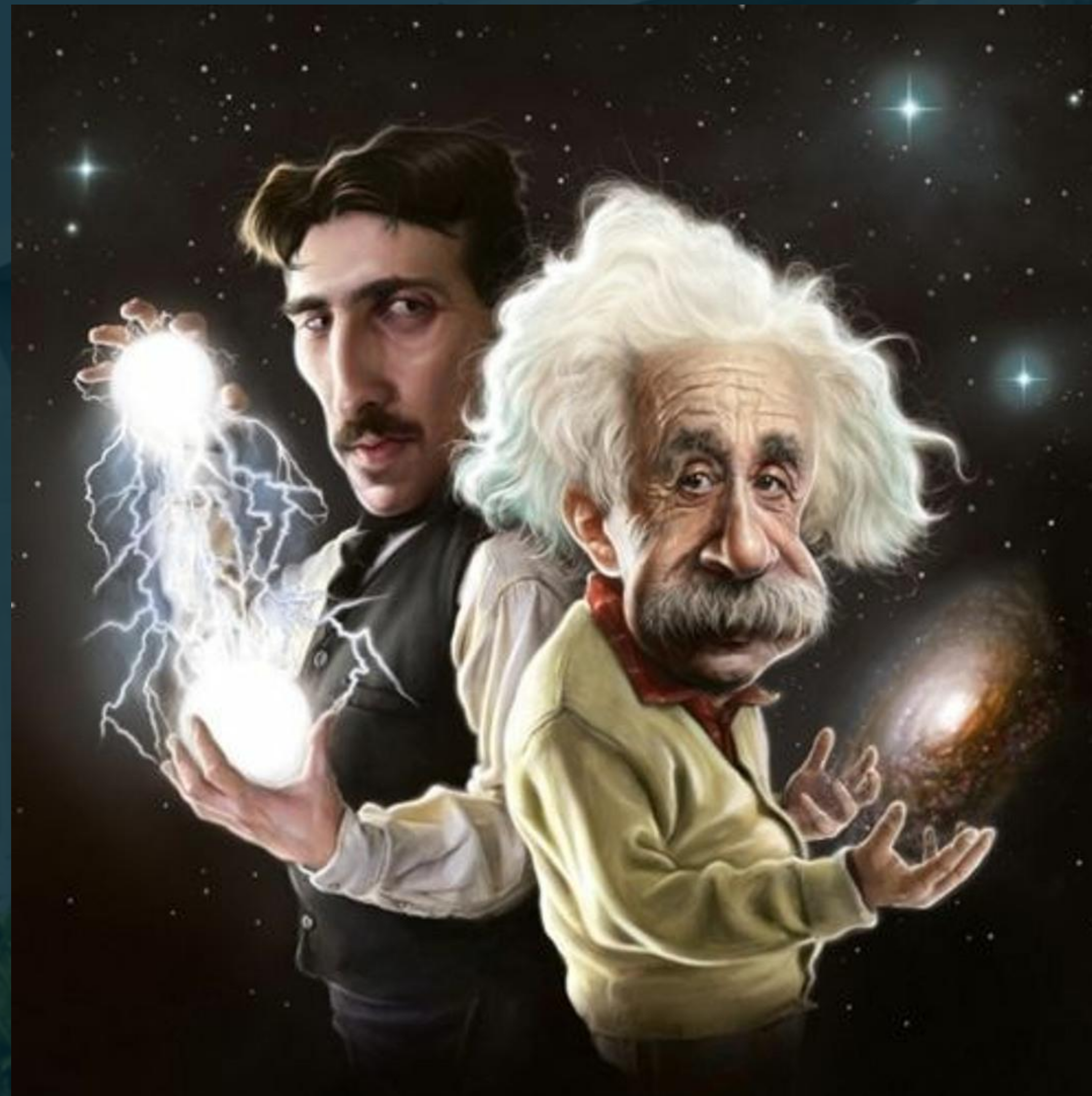
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Enjoy the IoT trip



Questions?

Thanks!

